

Ultrafast Solvation Dynamics and Photoinduced Electron Transfer in Functionalized Hexagonal Boron Nitride Quantum Dots: A Combined Spectroscopy and Quantum Chemical Study

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Abstract

Photoinduced charge separation and ultrafast solvation dynamics in zero-dimensional layered nanostructures play a fundamental role in the development of next-generation photocatalytic and optoelectronic devices. In this investigation, we elucidate the excited-state relaxation pathways and interfacial solvation dynamics of functionalized hexagonal boron nitride quantum dots (h-BN QDs) suspended in polar protic and aprotic solvent environments. Utilizing femtosecond broadband transient absorption spectroscopy spanning the spectral range of 380 to 750 nm alongside time-correlated single-photon counting (TCSPC) and hybrid density functional theory (DFT/TD-DFT) calculations, we isolate three distinct kinetic components governing the photoexcited state: an ultrafast inertial solvation response ($\tau_1 \approx 180\text{--}240$ fs), an intramolecular charge-transfer state formation ($\tau_2 \approx 2.4\text{--}3.8$ ps), and a slow diffusive solvent reorganization phase coupled with radiative recombination ($\tau_3 \approx 1.2\text{--}4.5$ ns). Solvent-dependent spectral shifts confirm a substantial increase

in the dipole moment upon optical excitation ($\Delta\mu \approx 8.4$ D), pointing to a strongly polarized charge-transfer state. Quantum chemical calculations support a localized electronic transition from oxygen-functionalized edge states to core boron nitride orbitals. These findings establish foundational physical chemistry principles for tailoring non-toxic 2D-based quantum dots in energy-conversion applications.

Introduction

Zero-dimensional materials derived from layered two-dimensional (2D) matrix systems have attracted widespread attention due to their unique quantum confinement effects, edge-state photoluminescence, and remarkable chemical stability.^{1,2} Among these emerging systems, hexagonal boron nitride quantum dots (h-BN QDs) serve as a wide-bandgap analog to graphene quantum dots, exhibiting exceptional chemical inertness, high thermal stability, and tunable photoluminescence spanning the ultraviolet to visible spectrum.³